

# SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1710

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## COURSE OUTCOME COVERED

CO3: Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)

Assessment Tool Weightage: 50%

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Time Duration: 1 Hour

QUESTIONS: 4

Total Marks:40

Annexure A

## CASE STUDY

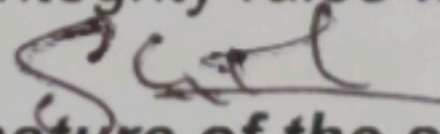
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Code of Conduct:

(92511347)

I Shanmuganathan (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

  
Signature of the student:

23+23+18+18+8

90

AP



## Question 1: Smart medical Diagnosis System

### 1) (a) Propositional Logic Representation

#### \* Propositional Symbols

- \* F: Patient has Fever
- \* C: Patient has Cough
- \* R: Patient has Rash
- \* B: Patient has Breathlessness
- \* F1: Possible Measles Flu
- \* M: Possible Measles
- \* P: Possible Pneumonia

#### \* Knowledge Base Rules & Facts

- \* Rule 1 (R1):  $F \wedge C \rightarrow F1$
- \* Rule 2 (R2):  $F \wedge R \rightarrow M$
- \* Rule 3 (R3):  $C \wedge B \rightarrow P$
- \* Facts:  $F = \text{True}, C = \text{True}, B = \text{True}$

### 1) (b) Forward Chaining

#### \* Initial known facts {F, C, B}

- \* Step 1: Check rules whose Premises are satisfied by known facts:

- \* Evaluate R1: Premises  $(F \wedge C)$  is  $\text{True} \wedge \text{True} = \text{True}$
- \* Fire R1: Derive F1 (Possible Flu).



- \* Step 2: Check remaining unsatisfied rules
- \* Evaluate  $R_3$ : Premises (CNB) is True  $\wedge$  True = false
- \* Fire  $R_3$ : Derive P (Possible Pneumonia)
- \* Update facts:  $\{F, C, B, F1, P\}$
- \* Step 3: Evaluates  $R_2$ : Premise (FNR) fails because R is unknown.
- \* Derived Diagnoses for patient A: Possible Flu (F1) and Possible Pneumonia (P)

1f)

Backward Chaining & Goal Reducing Tree

- \* Goal: Confirm P (patient A has Pneumonia)
- \* Step 1: Match goal P with rule consequents  
 $\rightarrow$  Rule 3 (CNB  $\rightarrow$  P) matches.
- \* Step 2: Sub goals Created: Prove C and Prove B
- \* Step 3: Check C in KB  $\rightarrow$  C is an initial fact (True).
- \* Step 4: Check B in KB  $\rightarrow$  B is an initial fact (True)
- \* Goal reduction tree verified: All Sub goals resolve to ground facts, confirming P is true.



\* Rule 1:

$\text{Vehicle}(x) \wedge \text{Emergency Type}(x) \rightarrow \text{clear path}(x)$

\* Facts:  $\text{Vehicle}(\text{Ambulance})$ ,  
 $\text{Emergency Type}(\text{Ambulance})$

\* Step 1: match  $\text{Vehicle}(x)$  with

$\text{Vehicle}(\text{Ambulance}) \Rightarrow \theta_1 = \{x / \text{Ambulance}\}$

\* Step 2: Substitute  $\theta_1$  into

$\text{Emergency Type}(x) \Rightarrow \text{Emergency Type}(\text{Ambulance})$

\* Step 3:

\* Match Emergency Type with fact

\*  $\text{Emergency type}(\text{Ambulance}) \Rightarrow Q_2 = \{\}$

\* Most General unifier (mgu):

$\theta = \{x / \text{Ambulance}\}$

2(b) Resolution Proof for  $\text{proceed}(\text{CarA})$

1) Convert Knowledge Base to CNF:

\*  $C_1$ :

$\neg \text{Vehicle}(x) \vee \neg \text{Emergency Types}(x) \vee \text{clear path}$

\*  $C_2$ :

$\neg \text{clear path}(y) \vee \text{Green Signal}(y)$

\*  $C_3$ :

$\neg \text{clear path}(y) \vee \neg \text{Red Signal}(y)$



\*  $C_5$ :

Vehicle (Ambulance)

\*  $C_6$ :

Emergency Type (Ambulance)

\*  $C_7$ :

Vehicle (Car)

\*  $C_8$ :

Behind (Car, Ambulance)

\*  $C_9$ :

(Negotiated Goal):  $\neg$  Proceed (Car)

## 2) Resolution Steps:

1) Resolve  $C_1$  with  $C_5$  (1/Ambulance):

$C_{10} = \neg$  Emergency (Ambulance)

2) Resolve  $C_{10}$  with  $C_6$

$C_{11} = \text{Clearpath (Ambulance)}$

3) Resolve  $C_2$  with  $C_{11}$  (4/Ambulance):

$C_{12} = \text{Green Signals (Ambulance)}$

4) Resolve  $C_4$  with  $C_9$  (2/Car)

$C_{13} = \neg$  Vehicle (Car)  $\vee \neg$  Behind (Car, Ambulance)  $\vee \neg$  Green signal

5) Resolve  $C_{13}$  with  $C_7$ :

$C_{14} = \neg$  Behind (Car, Ambulance)  $\vee \neg$  Green signal

6) Resolve  $C_{14}$  with  $C_8$

$C_{15} = \neg$  Green signal (Ambulance)



Empty class derived protocol (Carn) is proven true.

2(c) Analysis for two Simultaneously Emergency Vehicles.

\* Inadequacy: The Current KB lacks resources conflict resolution (eg: priority ranking or intersection clearance order)

\* Required Additions:

\* Priority predicate: Priority (x, y) to establish which emergency vehicle goes first

\* Mutual exclusion rule:

$clearpath(x) \wedge clearpath(y) \wedge Intersect(x, y) \rightarrow$  either  $x$  has higher priority.

Question 3: Agriculture Expert Reasoning System

3(c) Conversion to Conjunctive Normal Form (CNF)

1)  $P_1: SoilDry \rightarrow Irrigation\ Needed$

\* Eliminate  $\rightarrow$ :

$\neg SoilDry \vee Irrigation\ Needed$  (CNF)

2)  $P_2:$

$Irrigation\ Needed \wedge Crop\ Wheat \rightarrow Apply\ Drip\ method$

\* Eliminate  $\rightarrow$ :



(CNF2)

3 P3:

$\neg$  Apply Drip method  $\wedge$  Crop Whoot  $\rightarrow$  Crop at Risk

\* Eliminate  $\rightarrow$ :

$\neg$  ( $\neg$  Apply Drip method  $\wedge$  Crop Whoot)  $\vee$  Crop at Risk

move  $\neg$  inward

Apply Drip method  $\vee$   $\neg$  Crop Whoot  $\vee$  Crop at Risk

(CNF3)

4) Facts:

\* Soil Dry (CNF4)

\* Crop Whoot (CNF5)

3(b) Resolution Refutation for Goal

\* Negate Goal  $\neg$  Apply Drip method (CNF6)

Proof Tree Steps:

1) Resolve CNF1

2) Resolve CNF2

3) Resolve Clause B

4) Resolve clause C and CNF6



- \* without Soil dry: Irrigation Needed cannot be derived
- \* Consequently: Apply Drip method cannot be inferred via  $P_2$
- \* Does CropAtRisk follow? No. Statement  $P_3$  is an implication.

Failing to prove Apply Drip method does not logically assert  $\neg$  Apply Drip method. Therefore, CropAtRisk does not follow.

Question 4 Humpus World Knowledge Agent

4(a) Agent Belief State & Modus Ponens Derivation

\* Percept Facts: Stench [1,2], Breeze [1,1], Glitter [2,2]

\* Apply modus ponens

$$\left( \frac{\alpha \rightarrow \beta, \alpha}{\beta} \right)$$

1) Given

Stench [1,2]  $\rightarrow$  Humpus adjacent [1,2]  
and Stench [1,2]

$\Rightarrow$  Humpus is adjacent to [1,2]

2) Given Breeze [1,1]  $\rightarrow$  Pit adjacent [1,1]

and Breeze

$\Rightarrow$  Pit is



3) Given Gitter  $[2,2] \rightarrow$  Gold  $[2,2]$  and Gitter  $[2,2]$

$\Rightarrow$  Gold is at  $[2,2]$ .

4b) Possible Location of wumpus and pit

\* Adjacent to  $[1,2]$ : Adjacent grid cells are  $[1,1]$ ,  $[1,3]$ ,  $[2,2]$

\*  $\Rightarrow$  Wumpus is in  $\{[1,1], [1,3], [2,2]\}$ .

\* Adjacent to  $[1,1]$ : Adjacent grid cells are  $[1,2]$ ,  $[2,1]$

\*  $\Rightarrow$  Pits is in  $\{[1,2], [2,1]\}$ .

4c) Path Safety Evaluation

1) Cell  $[1,2]$  Safety check

\* Pit adjacent to  $[1,1]$  means a pit could be at  $[1,2]$  or  $[2,1]$

\* Since pit at  $[1,2]$  cannot be ruled out  
( $\neg$  pit  $[1,2]$  is unproven), Rule 9

2) Conclusion:

$[1,1] \rightarrow [1,2] \rightarrow [2,2]$  is not guaranteed safe

because both intermediate destination cells have